

Hein Zarni Aung

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EDUCATION

Stony Brook University

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Full Stack Development, Software Security, Programming Abstractions

Expected Graduation: May 2027

INTERNSHIPS

Software Engineering Intern — Clustr

Cross-platform React Native social/scheduling app | ~8-person dev team

Jan 2026 – Sept 2026 (expected)

- Built server-side backend infrastructure (Firebase Cloud Functions) for a React Native app, including CRUD operations, permission enforcement, and activity feed endpoints, powering features used by real end users.
- Designed and implemented a Scheduler UI module featuring tabbed navigation, a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.

Software Engineering Intern — Prenora Dynamics LLC

~8-person team

Jun 2026 – Sept 2026 (expected)

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, and built a seed-data workflow to streamline debugging and local development.

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Stony Brook University

PI: Abisheka Pitumpe | Team: 2

Sep 2025 – Dec 2025

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag potentially fraudulent domains impersonating EZ-Pass tolling services, using Selenium to automate scraping and Python to filter and triage flagged sites for manual review.

Vertically Integrated Project — Automotive Ethics Lab, Stony Brook University

Simulations sub-team (5 people) | Multi-semester, through graduation

Starting Fall 2026

PROJECTS

TuneAcademy

React, NumPy, Essentia, Firestore

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances, and integrating ElevenLabs for AI-generated voice feedback reports.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy to Cloudflare Workers, across a full stack spanning a React/Tailwind frontend and a Python/FastAPI backend in Docker.

CAMPUS INVOLVEMENT

Teaching Assistant - Data Structures & Algorithms, Stony Brook University

Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures.
- Created original review materials from scratch for midterm exam preparation, adopted as the basis for subsequent semesters' review sessions.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, C

Frontend: React Native, Next.js, React

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, Firebase/Firestore, MongoDB, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium